

5.2.1. Titanic Dataset

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Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:

The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the `pandas` library. It contains the following columns:

- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
- `Gender`: Gender of the passenger (male/female).
- `Age`: Age of the passenger.
- `Survived`: Survival status (0 = Did not survive, 1 = Survived).
- `Fare`: Ticket fare paid by the passenger.

Visualization:

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:

Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution):

- Create a bar plot to show the distribution of passengers across the different passenger classes (`Pclass`).
- Use the color `skyblue` for the bars.
- Title the plot as **"Passenger Class Distribution"**.
- Label the x-axis as **"Pclass"** and the y-axis as **"Count"**.

Pie Chart (Gender Distribution):

- Create a pie chart to display the distribution of male and female passengers.
- Use `lightblue` for males and `lightcoral` for females.
- Include percentages on the slices (use `autopct='%1.1f%%'`).
- Title the plot as **"Gender Distribution"**.

Histogram (Age Distribution):

- Create a histogram to visualize the distribution of passengers' ages.
- Use `lightgreen` for the bars with black edges (`edgecolor = 'black'`).
- Set the number of bins to **8** for the histogram.
- Title the plot as **"Age Distribution"**.

- Label the x-axis as "**Age**" and the y-axis as "**Frequency**".

Bar Plot (Survival Count):

- Create a bar plot to show the count of passengers who survived and those who did not, based on the `Survived` column.
- Use the colors `lightblue` for survivors (1) and `lightcoral` for non-survivors (0).
- Title the plot as "**Survival Count**".
- Label the x-axis as "**Survived (0 = No, 1 = Yes)**" and the y-axis as "**Count**".

Scatter Plot (Fare vs Age):

- Create a scatter plot to visualize the relationship between the `Fare` and `Age` of passengers.
- Use orange for the data points.
- Title the plot as "**Fare vs Age**".
- Label the x-axis as "**Age**" and the y-axis as "**Fare**".

Note: Refer to the displayed plot in the sample test cases for better understanding.

Code:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Load the Titanic dataset from the CSV file
```

```
df = pd.read_csv('titanic.csv')
```

```
# Set up the figure for 5 subplots
```

```
fig, axes = plt.subplots(3, 2, figsize=(12, 12))
```

```
# write the code..
```

```
#plot 1: Count of passenger by class
```

```
axes[0,0].bar(df['Pclass'].value_counts().index,
```

```

df['Pclass'].value_counts(),color='skyblue')

axes[0,0].set_title("Passenger Class Distribution")

axes[0,0].set_xlabel("Pclass")

axes[0,0].set_ylabel("Count")


#plot 2: Gender Distribution

axes[0,1].pie(df['Gender'].value_counts(),labels=df['Gender'].value_counts().index,autopct='%1.1f%%',c
olors=['lightblue','lightcoral'])

axes[0,1].set_title("Gender Distribution")


#plot 3: Age Distribution

axes[1,0].hist(df['Age'].dropna(),bins=8,color='lightgreen',edgecolor='black')

axes[1,0].set_title("Age Distribution")

axes[1,0].set_xlabel("Age")

axes[1,0].set_ylabel("Frequency")


#plot 4: Survival Count

axes[1,1].bar(df['Survived'].value_counts().index,df['Survived'].value_counts(),
color=['lightblue','lightcoral'])

axes[1,1].set_title("Survival Count")

axes[1,1].set_xlabel("Survived (0 = No, 1 = Yes)")

axes[1,1].set_ylabel("Count")


axes[2,0].scatter(df['Age'],df['Fare'],color='orange',edgecolors='black')

axes[2,0].set_title("Fare vs Age")

axes[2,0].set_xlabel("Age")

```

```
axes[2,0].set_ylabel("Fare")
```

```
plt.tight_layout()
```

```
plt.show()
```

